01 The Growth Room

Unlimited growth & its impossibility

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 31 min	**+ 28 min**	Nothing. This is the first room.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Growth tab

Every individual gives birth at rate b and dies at rate d, however crowded the world becomes. The dashed line is the textbook N_0e^{rt}; the solid lines are populations in which every birth and death is a random event. This guide asks what such a population does, and why it cannot exist for long.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain exponential growth as a constant per-capita rate $r = b - d$.
- Recognise exponential growth as a straight line on a logarithmic axis, and read a doubling time.
- Explain why a small population can go extinct even when $r > 0$.
- Explain why the per-capita growth rate is flat against density in this model, and why that is the room's key assumption.

The extension adds

- Use the extinction probability $(d/b)^{N_0}$ to predict how many replicates are lost.
- Show that two populations with the same r but different turnover (b and d) behave differently.
- Use the doubling time $\ln 2/r$ to make quantitative predictions.

SUGGESTED ROUTES

- **35-minute practical (BSc):** steps 1–4 and questions 1–3.
- **First session of a population ecology course:** this room and the Crowding Room core paths together, about 70 minutes.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 The exponential curve

8 min

SET UP

The defaults: N_0 = 10, b = 0.50, d = 0.30, time span 40, 1 replicate, linear axis.

PREDICT FIRST

What is r? Roughly how long does it take the population to double? Sketch the curve you expect.

RUN

Press **Run**. Then switch the vertical axis to **Logarithmic**.

LOOK AT

the readouts $r = b - d$ and *Doubling time*, and the shape of the curve on each axis.

EXPLAIN

Why does the curve become a straight line on the log axis? What does its slope represent?

2 Barely growing

10 min

SET UP

b = **0.31**, d = 0.30, N_0 = **5**, **20** replicates.

PREDICT FIRST

The average population should grow very slowly. Out of 20 replicates, how many do you expect to go extinct?

RUN

Press **Run**.

LOOK AT

the solid lines that fall to zero, against the dashed prediction.

EXPLAIN

How can a population whose expected growth is positive go extinct? Which kind of population is most at risk?

3 Two populations, one difference

8 min

SET UP

Back to the defaults. *Compare two populations*: **On**. Population 2 has $d = 0.40$.

PREDICT FIRST

Population 2's r is half of population 1's. How will the two compare after 40 time units: half as many, or far fewer than half?

RUN

Press **Run** on a linear axis, then on a log axis.

EXPLAIN

Why does a small difference in rate turn into a huge difference in numbers?

4 The flat line

5 min

SET UP

The run you just made.

PREDICT FIRST

What will the per-capita growth rate look like plotted against density?

LOOK AT

Per-capita growth rate vs. density and *Total growth rate vs. density*.

EXPLAIN

What would have to change about that flat line for the population to stop growing? (This is the question every other room answers.)

EXTENSION · MSC

5 Predicting extinction

10 min

SET UP

One population, 20 replicates, $d = 0.30$. $b = 0.35$ with $N_0 = 5$, then $N_0 = 20$.

PREDICT FIRST

For a birth–death process the chance of eventual extinction is $(d/b)^{N_0}$. Calculate it for both starting sizes, and the number of the 20 replicates you expect to lose.

RUN

Run each setting twice and count the extinctions.

EXPLAIN

Why does the starting size matter so much, and what does that imply for reintroductions?

6 Same r, different turnover

10 min

SET UP

Compare two populations, 20 replicates, N_0 = 10 in both. Population 1: b = 0.50, d = 0.30. Population 2: b = **0.90**, d = **0.70**.

PREDICT FIRST

Both have r = 0.2. Will they behave the same? Calculate the extinction probability for each.

RUN

Run twice.

EXPLAIN

Why does the population with faster turnover show more scatter and more extinctions, even with the same expected growth?

7 How long until it runs off the chart?

8 min

SET UP

The defaults, linear axis, time span **60**.

PREDICT FIRST

The linear chart is drawn up to 200,000 individuals. Using N_0e^{rt}, when should the population pass that line?

RUN

Run and note when the line leaves the top of the panel.

EXPLAIN

Calculate how many individuals the model predicts after 200 time units. Why is that answer absurd, and what does that tell you about exponential growth in nature?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc A population of rats on an island grows from 10 to 160 in four years. If growth is exponential, what is the doubling time and what is r?
2. BSc Why is exponential growth the right description of a population at the start of an invasion, but never for long?
3. BSc Explain to a conservation manager why a reintroduced population of five animals with a positive growth rate might still fail.
4. MSc Demographic stochasticity is distinct from environmental stochasticity. Define both and say which this room includes.
5. MSc Two populations have the same r but different b and d. Which would you rather manage as an endangered species, and why?
6. MSc Human population growth is sometimes described as “faster than exponential”. What would that mean in terms of r?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** $r = 0.2$, doubling time $\ln 2/0.2 \approx 3.5$ time units. On a log axis the curve is a straight line with slope r.
- **Step 2.** Most replicates go extinct: the chance for each is $(0.30/0.31)^5 \approx 0.85$. Small populations are at the mercy of individual births and deaths.
- **Step 3.** Population 2 ends far below half of population 1: the gap grows as $e^{(0.2 - 0.1)t}$, about 55-fold after 40 time units.
- **Step 4.** A flat line: an individual does equally well at any density. Only a per-capita rate that falls with density can stop growth.
- **Step 5.** $(0.30/0.35)^5 \approx 0.46$, so about 9 of 20; $(0.30/0.35)^{20} \approx 0.05$, so about 1 of 20.
- **Step 6.** Extinction probability is $(0.3/0.5)^{10} \approx 0.006$ against $(0.7/0.9)^{10} \approx 0.08$: more turnover means more randomness for the same r.
- **Step 7.** About 50 time units ($t = \ln 20{,}000/0.2 \approx 49.5$). After 200 time units the model predicts about 2.4×10^{18} individuals: no real environment allows it.